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RESEARCH-ARTICLE

A Systematic Literature Review Study of Digital Transformation in Higher Education: Technologies, Challenges and Trends

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A Systematic Literature Review Study of Digital Transformation in Higher Education: Technologies, Challenges and Trends

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Abstract

The digital transformation of higher education is occurring rapidly around the globe, leaving an indelible mark on improving the quality of education, expanding the accessibility of educational resources, and realizing innovation and reform in teaching and research. The extant research on digital transformation in higher education is relatively limited in scope and has certain inherent limitations. Therefore, this paper employs the method of systematic literature review (SLR) to provide a comprehensive analysis of the digital transformation of higher education from three perspectives: digital technology, existing challenges, and future trends. This analysis aims to identify potential correlations between these three dimensions and offer valuable insights for those engaged in the education industry and researchers. Furthermore, this study seeks to deepen our understanding of digital transformation and to provide guidance for future research directions.

CCS Concepts

• General and reference; • General conference proceedings;

Keywords

Digital Transformation, Higher Education, Digital Technology, Challenges, Future Trend

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1 Introduction

In the current era of globalisation and the rapid development of information technology, digitisation has become an essential strategy for enhancing competitiveness and innovation in various industries. However, it has also attracted the attention of many academic researchers and has attracted their interest in the field. Digitisation is converting information and processes into digital form using digital technologies [1]. Digital transformation is the systematic transformation of work patterns and processes through the deep integration of digital technologies to achieve greater efficiency and innovation [2].

The digital transformation in education is not just the introduction of technological tools but also a complete redesign of the teaching process, management model, and educational experience. Among these, educational technology management combines educational theory and technology application, using various tools, resources and technologies to serve teachers' teaching and students' learning [3]. The emergence of virtual classrooms and online learning platforms such as MOOCs has changed the traditional teaching mode and placed higher demands on teachers' skills [4]. At the same time, the rapid development of technologies such as the internet of things (IoT), artificial intelligence (AI), and blockchain has provided solid support for the management of educational technology and significantly improved its efficiency [5]. Therefore, digital transformation has not only brought about new modes and methods of teaching and learning to improve the quality of education but has also had a profound impact on educational technology management.

The digitalisation of higher education has become a hot topic for educators, Timotheou et al. exploring the impact of digital technologies on the digitalisation of higher education [6], Bong and Chen discussing the competencies of teachers in the context of the digitalisation of education [7], and Truong and Diep exploring the technological highlights and trends of the current digital transformation of higher education [8]. Bisri et al. used the SLR to identify the critical success factors in the digital transformation process in higher education [9]. Zhou et al. analysed the factors influencing how to recommend digital educational resources accurately based on user characteristics [10]. Although these papers have examined the key issues in digital transformation from different perspectives, the relatively singular nature of the research object and the lack of

Table 1: Inclusion and exclusion criteria for reference selection

Inclusion Criteria		Exclusion Criteria	
IC1	Related to the topic of digital transformation research in higher education	EC1	Papers published before 2020
IC2	Including, but not limited to, papers related to the application of digital technologies in higher education	EC2	Conference papers, books, magazines, short papers, technical reports, literature reviews or tutorials
IC3	Peer-reviewed papers	EC3	Not written in English
		EC4	Not address the use of digital technology in higher education

a comprehensive perspective to analyse, which has certain limitations. This paper comprehensively explores the research topic of digital transformation of higher education from the aspects of digital technology, current challenges and future development trends. Therefore, this paper formulates the following research questions.

RQ1: Which digital technologies play an important role in the digital transformation of higher education?

RQ2: What are the current challenges in the digital transformation process in higher education?

RQ3: What is the future of digital transformation in higher education?

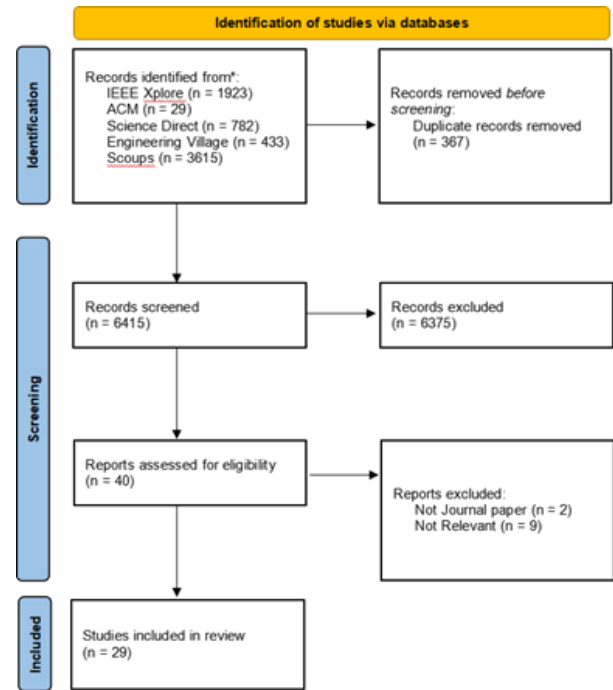
The remainder of the paper is structured as follows: Section II focuses on the research methodology and process used in this paper. Immediately after that, the study results are presented in Section III. The results are then discussed and analysed considering the research questions posed. In Section IV, the conclusions of this paper are given, as well as an indication of future research directions.

2 Methodology

This paper presents a systematic review and analysis of relevant research topics over the last five years using the most common research methodology for SLR, based on the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA), a set of standardised guidelines for reporting systematic reviews and meta-analyses, including a 27-item checklist and a four-stage flowchart to help researchers ensure that their studies are of high quality and reproducible for peer review and application [11].

This review covers published research theories and papers on digital transformation in higher education over the last five years. Referring to the experience of researching previous papers, relevant searches were conducted in IEEE Xplore, ACM, ScienceDirect, Engineering Village and Scoups, a comprehensive academic database [6] [9]. Prior to the search, the specific keywords "digitalisation", "digital transformation", "digital technolog*", "higher education", "tertiary education" was combined with the logical operators "AND" and "OR". The search string is: ("digital transformation" OR "digitalisation" OR "digital technolog*") AND ("higher education" OR "tertiary education"). These terms were used to search for categories such as titles and abstracts of papers to gain an initial understanding of the research topic.

After an initial search in the digital library, 6782 papers were retrieved (1923 from IEEE Xplore, 29 from ACM, 782 from ScienceDirect, 433 from Engineering Village and 3615 from Scoups). After

**Figure 1: The PRISMA flowchart of reference screening.**

removing duplicates, 6,415 papers remained. 6375 papers were excluded by reading the titles, abstracts and keywords of these papers and screening them against the inclusion and exclusion criteria in Table 1. The 40 papers selected after the initial screening were then carefully read and evaluated, 29 papers were identified as meeting the relevant requirements and included in the further review. Figure 1 illustrates the specific reference screening process.

3 Results and discussion

The significance of digital transformation in higher education can be attributed to its capacity to not only transform the conventional approach to education but also to engender far-reaching implications for the education system [12]. Following a comprehensive examination of 29 papers, this paper will analyse the issue of digital transformation in higher education from digital technology, challenges associated with technology and future trends.

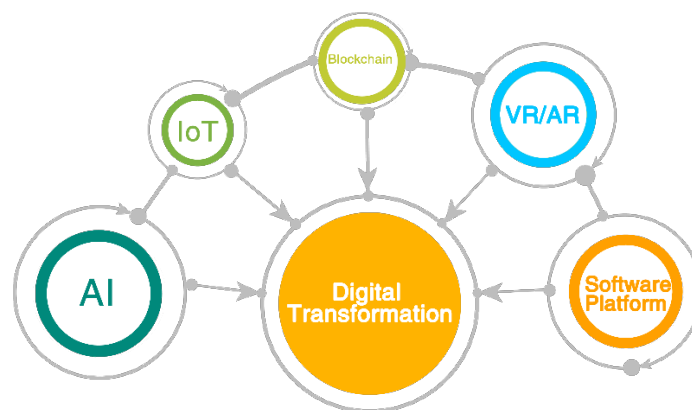


Figure 2: The digital technologies driving digital transformation in higher education.

3.1 Digital Technologies in the Digital Transformation of Higher Education (RQ1)

As illustrated in Figure 2, AI, IoT, Augmented Reality (AR) and Virtual Reality (VR), Blockchain, software and platforms have been instrumental in propelling the digital transformation over the past five years. AI, as a research focal point and a nascent professional domain in recent times, is progressively assuming a pivotal role in this transformation.

3.1.1 Artificial Intelligence. AI technology is employed across numerous sectors, including healthcare, finance, manufacturing, retail, and education. AI technology enhances the efficacy of teaching and the quality of learning and expedites the digital transformation process of higher education through personalised learning, automated management, and intelligent assessment. Table 2 presents a synthesis of the application of AI in digital transformation.

The advent of generative artificial intelligence (GAI), exemplified by ChatGPT, is transforming the landscape of higher education, offering invaluable assistance to educators and learners alike. GAI's capabilities extend beyond mere automation; it provides personalised learning content generation, intelligent assessment and feedback, enhancing the efficiency and quality of education while creating avenues for students and faculty to innovate and explore [13]. In a pioneering study, Keiper et al. delved into the potential of ChatGPT in sports management education, evaluating its generated response quality and its suitability and limitations within this domain [14]. Michalon and Claudia discussed the effective integration and use of AI tools such as ChatGPT in higher education environments to enhance student's learning experience and improve their skills [15]. Pham et al. assessed the efficacy of ChatGPT by using it in an engineering education programme, including its benefits, challenges and impact on teaching and learning [16]. Salifu et al. explored the key factors influencing economics students' behavioural intention and actual use of ChatGPT in Ghanaian higher education [17].

In addition to GAI playing an essential role in digital transformation, there are also technologies such as machine learning and deep learning that are involved in the digital transformation process. Yeslyamov also explores the practical application and effectiveness

of software robots based on AI technologies as "robot teachers" in universities [18]. Furthermore, there are applications of AI algorithms that significantly contribute to the reform of teaching and improve its quality [19] [20] [21] [22]. Based on data mining techniques, Liang's pathway study effectively improves the quality of teaching and learning in school physical education [23].

3.1.2 Internet of Things. The IoT represents a significant opportunity for enhancing teaching and digital transformation in higher education by integrating physical devices, sensors and the Internet [28]. Implementing IoT technology in educational institutions enables the creation of smart campuses, thereby facilitating the development of a more conducive and efficient learning environment for teachers and students [29]. Furthermore, integrating IoT in the construction of smart classrooms enables the utilisation of intelligent teaching tools by enhancing interconnectivity between teaching equipment. For instance, experimental equipment equipped with IoT sensors can monitor students' operating data in real-time and automatically provide this information to teachers, facilitating the timely adjustment of teaching content and methods. The collection and analysis of real-time data not only improves teaching accuracy but also enables teachers to identify students' learning needs and difficulties, thus facilitating the provision of more personalised tutoring [8]. Haque et al. investigate IoT's potential to transform traditional education systems into more flexible e-learning systems [30].

3.1.3 Augmented Reality and Virtual Reality. AR and VR technologies provide powerful tools for innovation in teaching and learning in higher education, driving the digital transformation of education. AR and VR enable students to understand complex concepts and theories more intuitively by creating immersive and interactive learning environments [31]. Kim et al. found that immersive virtual reality (IVR) content using 360-degree video was more effective than traditional 2-D video instructional methods in increasing student engagement, immersion, motivation, cognitive benefits and perceived learning outcomes than traditional 2-D video teaching methods [32]. Yang and Liu found that IVR teaching methods not only improved learners' skill transfer but also that learning

Table 2: The application of AI technology in the digital transformation of higher education

Citation	Research Issue	Technology
[14]	An investigation of the effectiveness of ChatGPT in sport management education and its possible educational impact.	ChatGPT
[15]	Effective use of AI tools such as ChatGPT to enhance students' critical thinking and communication skills.	ChatGPT
[16]	Advancing the digital transformation of engineering education using AI-assisted learning.	ChatGPT
[17]	To explore what factors influence the behavioural intentions and actual use of ChatGPT by economics students in higher education.	ChatGPT
[24]	A digital technology-enabled hybrid English online and offline teaching model.	Deep Learning
[25]	Explored the application of smart teaching in the digital transformation of English learning by integrating deep learning technology.	Deep Learning
[19]	Explored how digital technology can be used to improve teaching and learning.	AI algorithm
[18]	To explore the practical application and effectiveness of software robots based on artificial intelligence technology as "robot teachers" in universities.	AI Robots
[20]	How to apply AI to multimedia information processing and art and design teaching in order to improve teaching effectiveness.	AI algorithm
[26]	Exploring how Web 3.0 and AI technologies are affecting the development and reform of accounting education.	AI/Web3.0
[23]	Improving the quality of physical education and teaching in schools through data mining techniques.	DataMining
[27]	Intelligent scoring of English essays using machine learning to improve assessment efficiency and accuracy.	Machine Learning
[21]	Optimising teaching methods and evaluation systems for visual communication design in the perspective of digital technology.	AI algorithm
[22]	How to build a smart teaching platform for English in colleges and universities based on deep learning to improve the teaching effect.	AI algorithm

Table 3: The application of AR/VR in the digital transformation of higher education

Citation	Research Issue	Technology
[32]	Exploring the impact of immersive virtual reality content using 360-degree video on learning outcomes in undergraduate education.	VR
[34]	Ways in which AR can improve collaborative and self-directed learning in engineering education.	AR
[35]	Using digital experiential learning to enhance sustainable horticulture and landscape management education.	AR/VR
[36]	Effective application of digital management paradigms and emerging technologies in construction education.	VR
[37]	Implications of AR for socio-education in a sustainable learning ecology.	AR
[38]	How WebVR can be used to transform the teaching and learning experience in higher education.	WebVR
[33]	How IVR Impacts Learner Skill Transfer	IVR

engagement played a significant moderating role in the process [33].

In addition, AR technology can overlay digital information on real-world scenarios to help students better connect theory and practice. Jovan et al. found that AR technology in engineering education significantly facilitated students' understanding of pneumatic control schemes and helped them achieve desired learning outcomes faster [34]. Kee and Zhang's study showed that using AR and VR technology for tree management improved horticultural and landscape management learning for environmental sustainability and practical effectiveness [35]. Pugacheva et al. promoted students' understanding of complex construction processes and practical skills through virtual technology in construction education [36].

Furthermore, Gómez-Galán et al. contend that AR can significantly enhance socio-educational teaching in sustainable learning environments [37]. Additionally, AR and VR technologies can facilitate novel avenues for distance learning. Students can engage in a virtual classroom through VR, interacting with the teacher and their peers in a natural environment. Glasserman-Morales et al. demonstrate that using WebVR tools in higher education can markedly contribute to advancing disciplinary and transversal competencies [38]. Table 3 shows the result.

3.1.4 Blockchain. Blockchain technology, as a decentralised, tamper-proof distributed ledger, offers innovative solutions for higher education's pedagogical and digital transformation [39]. Blockchain technology enables educational institutions to achieve

Table 4: The application of software and platform in the digital transformation of higher education

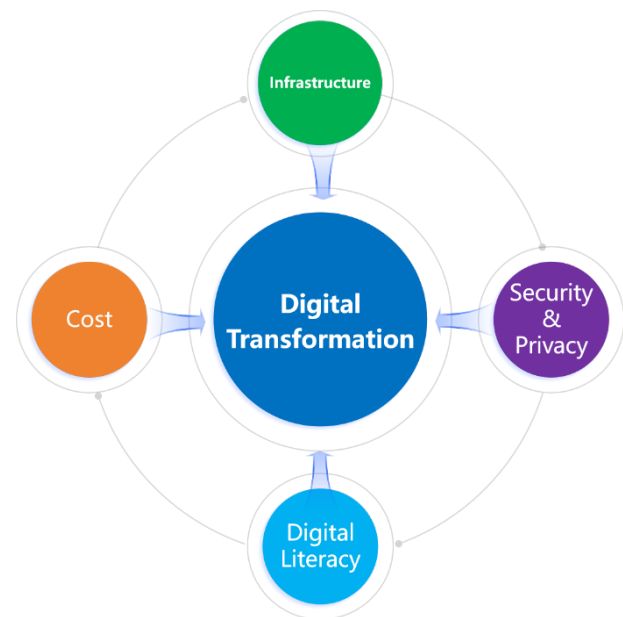
Citation	Research Issue	Technology
[42]	To design and apply cloud-based embedded voice-enabled teaching and learning systems to enhance the digital transformation of higher education.	Cloud Platform
[44]	Effectiveness and feasibility of using WeChat apps to enhance biosafety education in medical laboratories.	WeChat
[45]	The impact of WhatsApp on student performance and team effectiveness in a higher education learning environment.	WhatsApp
[46]	The potential of WhatsApp in expanding online teaching and learning for undergraduate students in South African universities.	WhatsApp
[47]	An approach to designing serious games to enhance learning in manufacturing system configuration and analysis in engineering education.	Serious Games
[48]	Support teaching and learning with dynamic and interactive tools to increase student engagement and understanding.	Software
[49]	Effective use of technology-supported teaching and learning processes by educators during the epidemic.	MOOCs

transparent and automated management of qualifications, transcripts and certificates. The immutable nature of blockchain data allows for the secure and reliable storage and sharing of student's academic records, preventing forgery or tampering [40]. Blockchain technology can facilitate the creation of personalised records of learning outcomes and lifelong learning. Kuleto et al. posit that the features of blockchain technology can be leveraged to help create a more sustainable education system [22]. Vatulkina and Cardoso proposed and implemented a certificate management system based on blockchain, digital signature, and QR code technologies to ensure certificate authenticity and tamper resistance and streamline the certificate validation process at Vinh University [41].

3.1.5 Software and Platform. The advent of cloud, mobile social platforms, and software applications has opened up new avenues for teaching and digital transformation in higher education. Conversely, cloud provide robust infrastructural support for educational institutions [8]. Li and Wu propose a cloud-based embedded speech teaching system that markedly enhances speech recognition's precision and noise resilience through the integration of neural networks [42].

Social platforms facilitate communication between students and teachers, enabling classroom discussion, project collaboration, and knowledge sharing [43]. These platforms foster community and interactivity in learning. Li et al. integrated an experiential learning model and a WeChat mini program to educate students about laboratory biosafety, demonstrating a notable enhancement in students' biosafety knowledge [44]. WhatsApp has also been employed as a prospective teaching and learning tool in higher education [45] [46].

Dedicated educational software offers the possibility of personalised learning. The successful development of a game for manufacturing system design and analysis through the application of the Serious Game Design framework by Urgo et al. has been demonstrated to enhance student engagement and learning outcomes significantly [47]. In [48], it was proposed that dynamic and interactive tools can significantly enhance students' comprehension and engagement with mathematical concepts, particularly in distance and classroom learning environments. Okoye et al. effectively


Figure 3: Challenges in the digital transformation of higher education.

employed MOOC technology to support the teaching-learning continuum in higher education during and after the pandemic [49]. Table 4 presents the details.

3.2 Challenges in the digital transformation of higher education (RQ2)

The various challenges posed by technology cannot be ignored in the process of digital transformation of higher education. Figure 3 illustrates the details.

3.2.1 Infrastructure. Infrastructure development is the cornerstone of digital transformation in higher education, determining the availability and effectiveness of digital learning environments. Efficient network connectivity, advanced hardware devices, robust learning

management platforms and powerful computing capabilities are all essential elements for the delivery of online education, virtual classrooms and personalised learning. For example, IoT involves the real-time connection of many devices and the transmission and processing of large amounts of data, so extensive and stable network coverage is key to ensuring the efficient operation of the system. By continuously improving and upgrading infrastructure, we can provide solid support for the digital transformation of higher education and promote improvements in educational quality and equity.

3.2.2 Cost. The high cost of infrastructure development is also one of the challenges universities faces in the digital transformation process. Digital transformation requires significant capital investments, including infrastructure upgrades, software purchases, hardware equipment updates, and technology training for faculty and students. However, many higher education institutions face budget constraints, making it difficult to rapidly deploy the necessary technologies and systems, thus slowing down the digital transformation process.

3.2.3 Security & Privacy. Data security and privacy are issues that cannot be overemphasised. Ensuring information security and privacy is the foundation for building digital trust, and only in a secure digital environment can an efficient and sustainable digital transformation of education be genuinely realised. Much sensitive information needs to be better protected, especially with the widespread use of online learning platforms and mobile social software.

3.2.4 Digital Literacy. The combined ability of individuals to use technological tools effectively in a digital environment, understand information, think critically, ensure data security, and create and communicate through digital means is known as digital literacy [50]. With good digital literacy, teachers can use technology effectively for innovative teaching and developing digital course content. At the same time, students will be able to access, evaluate and use digital resources more independently. Conversely, a lack of digital literacy will limit the effective use of technology and hinder improving the quality of education.

3.3 The Future of Digital Transformation in Higher Education (RQ3)

From 2020, there has been a notable increase in research activity surrounding the digital transformation of higher education. The advancement of digital technologies and the accelerated uptake of digital tools in higher education, particularly in the global response to the pandemic, have contributed to this growing interest. Figure 4 visually represents this trend, emphasising the growing significance of digital transformation in higher education and the necessity for further investigation.

Figure 5 illustrates the significant role of AI technology in driving the digital transformation of higher education. Moreover, it aligns with the growing prominence of AI as a research area and an emerging field of study. Therefore, AI can be posited to be a pivotal driver and trend in reforming higher education teaching and systems in the future. Secondly, AR/VR technology has also revolutionised pedagogical approaches and the learning experience.

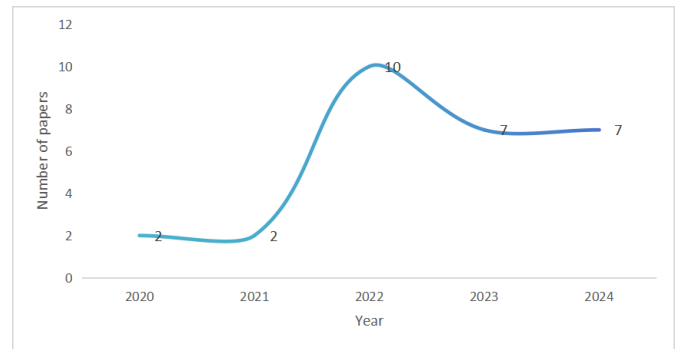


Figure 4: Trend of the number of papers reviewed from 2020-2024.

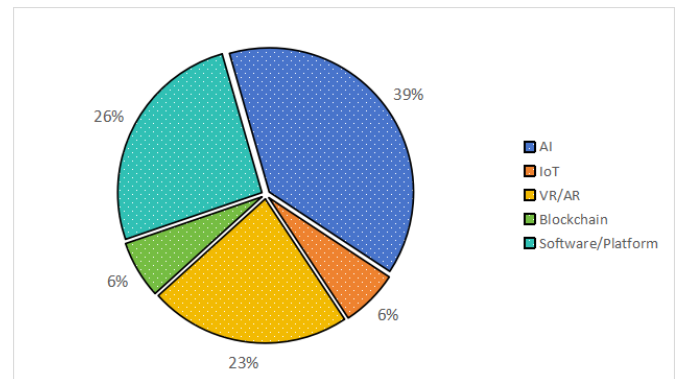


Figure 5: The percentage of various digital technologies in the literature on digital transformation in higher education.

The integrated deployment of cloud computing, mobile social platforms and software technologies has similarly transformed teaching methods, learning styles and all aspects of educational management, markedly enhancing the level of flexibility, interactivity and personalisation in higher education and driving a comprehensive digital transformation of education.

4 Conclusion

In the digital transformation of higher education, technologies such as AI, IoT, AR/VR, blockchain, cloud computing, mobile technologies, social platforms and educational software have played a key role in facilitating educational technology management. However, this transformation still faces several challenges, including inadequate infrastructure, high costs, security, privacy and lack of digital literacy. Higher education will become increasingly intelligent, personalised and equitable as technology advances. This paper aims to explore the impact of these technologies on higher education, analyse the challenges faced and examine future trends to provide educators and administrators with valuable guidance for more comprehensive and sustainable educational reform.

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